

9 Writing the code and Programming

After making the circuit and body, the next step is to write the code for the microcontroller. Since we have had a chapter on programming using C++ language, the code writing would not be difficult now.

9.1 Writing Code

A sample code for the line follower is as follows:

```
#define le pin_a0    //defining pin a0 to work as
input for left sensor, s1
#define re pin_a1    // defining pin a1 to work as
input for left sensor, s2
#define lm pin_a2    //s3not used at the moment, can
be used for improving efficiency later on
#define rm pin_a3    //s4, not used at the moment, can
be used for improving efficiency later on
#define FWD 0x05// 01 01 both motors in forward direction
#define RIGHT 0x04  // 01 00 left motor forward and
right motor stops
#define LEFT 0x01   // 00 01 left motor stops and
right motor forward
#define REV 0x0A // 10 10 both motors in rev direction
void main()
{SET_TRIS_A( 0xFF );           // FF=1111 1111, which
means we declare PORT A to work as input
  SET_TRIS_B( 0x00 );           // 00 = 0000 0000, which
means we declare PORT B to work as output
  while(1) //infinite loop, so that microcontroller keeps
on checking conditions repeatedly
  {
    if(le==0 && re==0)    // both sensors detect white
surface, so straight path ahead
    {
      output_b( FWD); // straight path ahead, so move forward
    }
    if(le==1 && re==0)    // left sensor detects black,
right sensor detects white, so left turn ahead
```